

VINOD AKSHAT

NIT Srinagar, Jammu and Kashmir, India

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Education

National Institute of Technology Srinagar

Bachelor of Technology in Computer Science and Engineering

August 2023 – May 2027

Srinagar, Jammu and Kashmir

Experience

Indian Institute of Technology Jammu

Research Intern, Cybersecurity

December 2025 – March 2026

Jammu, Jammu and Kashmir

- Reverse-engineered the **AFHDS-2A** UAV protocol from 30+ GB of RF captures (HackRF One, GNU Radio): built a GFSK demodulation chain (decimation, filtering, quadrature demod, Mueller–Müller clock recovery) and co-built a Lua dissector parsing bind-phase packet structure against live captures; hop-table extraction ongoing.
- Evaluated and reproduced battery-drainage (DoB) and DoS attack vectors on UAVs across multiple tools (**MAVProxy**, **MAVSDK**, **boofuzz**), and validated GPS-spoofing/failsafe behavior using **ArduPilot SITL** and **QGroundControl**.

Anantwave Quantum Computing

Backend Development Intern

July 2025 – October 2025

Andhra Pradesh, India

- Built an authentication backend (**Node.js**, **Express**, **TypeScript**) with **OAuth 2.0** (GitHub/Google), **Auth.js**, **JWT**, **PostgreSQL**, and **bcrypt** hashing.
- Implemented **RBAC** middleware for admin/user routes; debugged a separate team's **Python** backend via structured logging and root-cause analysis.

Projects

Warrant – Agentic Authorization Engine | *TypeScript, PostgreSQL, React* | [Github](#) | [Live](#)

Razorpay Buildathon

- Deterministic policy engine for **AI agent commerce**: separates agent intent from authorization authority via signed spending warrants and cryptographic replay protection.
- Fastify** API + **PostgreSQL** + **Razorpay** integration + **React** dashboard; deterministic ALLOW/BLOCK verdicts recorded in a hash-chained auditable ledger.

Yonro – Terminal Text Editor | *Rust* | [Github](#) | [Blog](#)

November 2025 – Present

- Extensible terminal text editor: grapheme-aware Unicode editing, custom text buffer, syntax highlighting, and 20+ Vim-style modal commands on a layered core with pane abstraction and an async **Tokio** plugin runtime.
- 7-suite **Criterion** benchmark (33 metrics) in CI: 538 ns pane relayout, 3.46 µs multi-pane edits, 11.6 ms full session; wired into **GitHub Actions**.

Grandma's Launcher | *Java, Android Studio, TypeScript, Docker* | [Github](#)

April 2026 – Present

- Accessibility-first Android launcher for senior citizens: large touch targets, text-to-speech guidance, high-contrast UI; engineered as a default home replacement via Android's Launcher APIs.
- Dockerized caretaker backend + PWA with magic-link auth and SOS live-location sharing; RSA-2048 handshake benchmarked at 2.75 ms sign / 121 µs verify, all under the 16.6 ms frame budget.

Grimoire – Systems & ML Foundations | *C, C++, Rust, Python, Java* | [Github](#)

December 2024 – Present

- 20+ algorithms and cryptographic primitives: **B/B+ Trees**, **LSM Trees**, **Bloom Filters**, a slab-based memory allocator, **RFC 1035** DNS resolver, **HMAC**, **RSA-2048**; plus compiler fundamentals and ML from first principles (incl. a neural network from scratch in C) with a **LangChain** + **Gemini** agent.
- Custom B-Tree beat `std::BTreeMap` and a production crate on search (40ms vs. 46ms/102ms).

Technical Skills

Languages: Rust, TypeScript, C, C++, Python, Java, JavaScript, SQL, Bash

AI/ML & Security: LangChain, LangGraph, Gemini API, agentic workflows, GNU Radio, Wireshark, protocol reverse-engineering, cryptography, RBAC

Backend & Frontend: Node.js, Express, Fastify, PostgreSQL, Auth.js, OAuth 2.0, JWT, bcrypt, React, Next.js, Vite

Cloud & Tools: Azure, Docker, GitHub Actions CI/CD, Linux/Unix, Git, GDB, Android Studio, Postman, Criterion benchmarking

Achievements & Leadership

- Lead Moderator, DEFCON Srinagar** – 15-member team; event logistics, speaker outreach, CTF activities.
- Polaris Fellowship 2026 Finalist** (Top 1%) – selected among the top 100 from 10,000+ applicants.
- Open Source, Zed Editor** – 3+ merged PRs, issue triage, contributor to a production editor.